

Model Lab

Every prior, same slate — 2026-09-21

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

37,000

Held-Out Scores

Fitted on plate appearances before 2026-08-15, scored on the 37,000 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5655	1.41×	0.02878	0.13300	0.00240
KNN comps Prior is the pooled rate of the nearest comparables.	0.5873	1.50×	0.02877	0.13254	0.00422
SVR Support-vector regression on the same feature matrix.	0.5774	1.39×	0.02886	0.13419	0.00826
Random forest Forest regression; a learned similarity metric.	0.5790	1.50×	0.02885	0.13371	0.00743
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5873	1.50×	0.02877	0.13254	0.00422
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5873	1.50×	0.02877	0.13254	0.00422
Linear regression Ordinary least squares on the same features.	0.5838	1.34×	0.02884	0.13361	0.00754
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5866	1.41×	0.02885	0.13322	0.00741
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5851	1.36×	0.02883	0.13318	0.00727
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5781	1.47×	0.02888	0.13412	0.00846

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 245, knn 245, svr 245, rf 245, core 245, ensemble 245, linear 245, logistic 245, nn 245, xgb 245 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,050 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5033	1.16×	0.02297	0.11221	0.01087
KNN comps Prior is the pooled rate of the nearest comparables.	0.5677	1.05×	0.02305	0.11258	0.01298
SVR Support-vector regression on the same feature matrix.	0.5623	1.05×	0.02349	0.12062	0.02515
Random forest Forest regression; a learned similarity metric.	0.5651	1.05×	0.02353	0.11846	0.02547
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5677	1.05×	0.02305	0.11258	0.01298
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5677	1.05×	0.02305	0.11258	0.01298
Linear regression Ordinary least squares on the same features.	0.5700	1.05×	0.02357	0.11963	0.02423
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5719	0.63×	0.02359	0.11674	0.02607
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5722	1.05×	0.02345	0.11601	0.02289
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5636	0.84×	0.02373	0.12064	0.02827

Slate Agreement

Flat league · 4/6 shared **2/6 hit**

Pete Alonso ✓
Spencer Torkelson ✓
Riley Greene
Dillon Dingler
Kody Clemens
Hao-Yu Lee

SVR · 4/6 shared **2/6 hit**

Emmanuel Rodriguez ✓
Pete Alonso ✓
Riley Greene
Coby Mayo
James Wood
Kody Clemens

Core (KNN+SVR) · 6/6 shared **3/6 hit**

Pete Alonso ✓
Emmanuel Rodriguez ✓
Riley Greene
Spencer Torkelson ✓
Kody Clemens
Eduardo Valencia

Linear regression · 5/6 shared **3/6 hit**

Emmanuel Rodriguez ✓
Pete Alonso ✓
Kody Clemens
Riley Greene
Spencer Torkelson ✓
Coby Mayo

Neural net (torch) · 4/6 shared **2/6 hit**

KNN comps · 6/6 shared **3/6 hit**

13.4% **Pete Alonso** ✓ **15.2%**
12.5% **Emmanuel Rodriguez** ✓ **15.0%**
12.4% **Riley Greene** **14.5%**
12.2% **Spencer Torkelson** ✓ **14.4%**
12.2% **Kody Clemens** **14.2%**
11.9% **Eduardo Valencia** **13.8%**

Random forest · 4/6 shared **2/6 hit**

20.9% **Emmanuel Rodriguez** ✓ **19.3%**
15.4% **Pete Alonso** ✓ **15.5%**
14.7% **Riley Greene** **14.5%**
14.7% James Wood **14.3%**
14.6% **Kody Clemens** **14.3%**
14.1% Coby Mayo **14.2%**

Forest blend · 6/6 shared **3/6 hit**

15.2% **Pete Alonso** ✓ **15.2%**
15.0% **Emmanuel Rodriguez** ✓ **15.0%**
14.5% **Riley Greene** **14.5%**
14.4% **Spencer Torkelson** ✓ **14.4%**
14.2% **Kody Clemens** **14.2%**
13.8% **Eduardo Valencia** **13.8%**

Logistic regression · 4/6 shared **2/6 hit**

18.1% **Emmanuel Rodriguez** ✓ **20.9%**
15.3% **Pete Alonso** ✓ **15.7%**
14.1% Coby Mayo **15.1%**
14.1% **Riley Greene** **14.7%**
13.9% **Kody Clemens** **14.6%**
13.8% James Wood **14.0%**

James Wood

14.1%

Samuel Basallo

Riley Greene

13.8%

⚖️ Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. James Wood slate Washington Nationals · vs LHP · 30 HR / 598 PA	10.9%	13.6%	14.6%	14.3%	13.6%	13.6%	13.5%	14.0%	14.1%
9. Dillon Dingler Detroit Tigers · vs RHP · 27 HR / 590 PA	12.2%	13.4%	13.8%	13.2%	13.4%	13.4%	13.4%	13.6%	13.5%
10. Bryce Eldridge slate San Francisco Giants · vs RHP · 17 HR / 448 PA	11.5%	13.2%	12.9%	13.0%	13.2%	13.2%	12.2%	11.8%	12.4%
11. CJ Abrams Washington Nationals · vs LHP · 31 HR / 631 PA	11.3%	12.8%	12.8%	12.7%	12.8%	12.8%	12.2%	12.5%	12.4%
12. Samuel Basallo Baltimore Orioles · vs RHP · 18 HR / 397 PA	11.3%	12.8%	13.5%	13.8%	12.8%	12.8%	13.0%	12.9%	13.4%
13. Abimelec Ortiz Washington Nationals · vs LHP · 5 HR / 129 PA	10.0%	12.6%	12.3%	12.4%	12.6%	12.6%	12.4%	11.4%	12.9%
14. Kazuma Okamoto slate Toronto Blue Jays · vs RHP · 33 HR / 613 PA	10.7%	12.0%	12.8%	12.2%	12.0%	12.0%	12.0%	12.7%	12.3%
15. Dylan Crews Washington Nationals · vs LHP · 12 HR / 419 PA	11.3%	11.9%	12.1%	12.4%	11.9%	11.9%	12.1%	11.8%	12.1%
16. Dylan Beavers Baltimore Orioles · vs RHP · 7 HR / 340 PA	11.0%	11.9%	11.2%	11.3%	11.9%	11.9%	11.5%	11.1%	11.5%
17. Hao-Yu Lee Detroit Tigers · vs RHP · 10 HR / 335 PA	11.9%	11.9%	11.2%	12.7%	11.9%	11.9%	11.9%	11.6%	11.8%
18. Christian Encarnacion-Strand Baltimore Orioles · vs RHP · 10 HR / 198 PA	10.1%	11.7%	12.1%	11.7%	11.7%	11.7%	11.3%	11.1%	11.8%
19. Walker Jenkins Minnesota Twins · vs RHP · 3 HR / 71 PA	11.9%	11.4%	12.3%	11.4%	11.4%	11.4%	11.7%	10.0%	11.2%
20. Daylen Lile Washington Nationals · vs LHP · 22 HR / 634 PA	10.5%	11.2%	11.2%	11.0%	11.2%	11.2%	11.5%	11.7%	11.8%
21. Ryan Jeffers Minnesota Twins · vs RHP · 14 HR / 353 PA	10.2%	11.2%	12.0%	11.2%	11.2%	11.2%	11.3%	11.2%	11.7%
22. Brady House	10.6%	11.0%	10.3%	10.9%	11.0%	11.0%	10.8%	10.5%	10.6%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Jung Hoo Lee San Francisco Giants · vs RHP · 10 HR / 579 PA	10.2%	8.5%	8.1%	8.1%	8.5%	8.5%	8.2%	8.5%	8.2%
38. Turner Hill San Francisco Giants · vs RHP · 1 HR / 110 PA	9.8%	8.4%	8.9%	9.2%	8.4%	8.4%	8.6%	7.3%	8.0%
39. Grant McCray San Francisco Giants · vs RHP · 2 HR / 89 PA	9.5%	8.2%	8.5%	8.9%	8.2%	8.2%	8.6%	7.5%	8.2%
40. Shay Whitcomb San Francisco Giants · vs RHP · 4 HR / 108 PA	8.6%	8.2%	9.2%	9.2%	8.2%	8.2%	9.4%	8.7%	9.1%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.